



IILM UNIVERSITY, GREATER NOIDA, U.P.

School of Computer Science and Engineering

INTERNSHIP REPORT

ON

PYTHON PROGRAMMING

Submitted in partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

Submitted By

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Roll Number: CS-2341655

Batch: 2026-27

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CANDIDATE'S DECLARATION

I, **Prakhar Tripathi** do hereby declare that the internship report titled “**Python Programming**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Prakhar Tripathi

Roll No: CS-2341655

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CodSoft for providing me with the opportunity to undertake a virtual internship in Python Programming. The internship provides a practical environment to strengthen programming concepts through project-based tasks. I am thankful to the School of Computer Science and Engineering, IILM University, Greater Noida, for academic guidance and support. I also express my gratitude to my mentor, faculty members, family, and peers for their encouragement and guidance throughout the internship process.

Signature

Student Name: Prakhar Tripathi

Roll No: CS-2341655

Internship Offer Letter



INTERNSHIP OFFER LETTER

Date : 24/08/2026

ID: BY26RY232903

Dear **Prakhar Tripathi**,

We would like to congratulate you on being selected for the **"Python Programming"** virtual internship position with **"CodSoft"**. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **25 August 2026 to 25 September 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!



ISO Number - TSNUK79053

Sincerely

A handwritten signature in black ink that reads "Ruman Arshad".

Ruman Arshad
Program Manager



MSME Regd



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4. PROJECT DESCRIPTION

The internship project focuses on the practical application of Python programming through the development of a collection of small-scale, functional applications. The project was undertaken as part of the Python Programming Virtual Internship at CodSoft, where participants are required to complete project-based programming tasks and maintain their work in a dedicated GitHub repository.

The primary focus of the project is to apply fundamental Python programming concepts to real application scenarios. The assigned tasks cover different types of programming requirements, including task management, arithmetic operations, password generation, interactive gaming, and contact management. Through these applications, concepts such as conditional statements, loops, functions, lists, dictionaries, modules, randomization, input validation, and exception handling are applied in practical situations.

The project consists of the following applications:

- To-Do List – A task management application designed to allow users to create, view, update, delete, and manage tasks.
- Calculator – An interactive application that performs basic arithmetic operations based on user input.
- Password Generator – A utility designed to generate randomized passwords according to the specified length and character requirements.
- Rock-Paper-Scissors Game – An interactive game in which the user plays against a computer-generated choice using predefined game rules and randomization.
- Contact Book – A contact management application that provides functionality for adding, viewing, searching, updating, and deleting contact information.

The development process involves understanding the requirements of each task, designing the program flow, selecting appropriate data structures, implementing the required functionality, and testing the applications with different inputs. Input validation and error handling are incorporated to improve the reliability and usability of the programs.

The project also provides practical exposure to basic software development practices such as code organization, debugging, testing, documentation, and version control. The source code of the completed internship tasks is maintained in a dedicated GitHub repository in accordance with the CodSoft internship requirements.

Overall, the project provides a practical environment for applying Python programming knowledge to functional applications. The combination of multiple projects helps develop programming logic, problem-solving ability, data-handling skills, debugging techniques, and familiarity with organizing software projects in a professional manner.

4.1 Introduction

Python is a high-level, versatile, and widely used programming language known for its simple syntax, readability, and extensive standard library. It is used across various domains of software development, including application development, automation, data processing, artificial intelligence, web development, and scripting. Due to its clear programming structure and extensive community support, Python is also an effective language for developing practical solutions while strengthening fundamental programming and problem-solving skills.

The Python Programming Internship at CodSoft provides a project-oriented environment for applying programming concepts to practical applications. The internship focuses on developing functional Python-based applications and gaining hands-on experience through a series of assigned tasks. The primary objective is to bridge the gap between theoretical programming knowledge and practical implementation by encouraging the development, testing, and refinement of working applications.

As part of the internship, a collection of application-based projects is undertaken, including a To-Do List, Calculator, Password Generator, Rock-Paper-Scissors Game, and Contact Book. Each project focuses on a different aspect of Python programming and contributes to the development of structured problem-solving skills. The To-Do List and Contact Book emphasize data management and CRUD-style operations, while the Calculator focuses on functions, arithmetic processing, and input validation. The Password Generator demonstrates the use of character sets and randomization, whereas the Rock-Paper-Scissors Game applies conditional logic, user interaction, and random computer selection.

The development of these applications involves the use of fundamental Python concepts such as variables, data types, conditional statements, loops, functions, lists, dictionaries, modules, user input, and exception handling. Particular emphasis is placed on creating programs that are organized, understandable, and capable of handling common user-input errors. The projects also provide an opportunity to understand how individual programming concepts can be combined to create complete, interactive applications.

The internship further introduces practical aspects of software development, including code organization, testing, debugging, documentation, and version control. A dedicated GitHub repository is maintained for the internship tasks, providing a centralized platform for organizing and presenting the developed projects. This approach encourages better coding practices and provides experience in maintaining software projects in a professional development environment.

Overall, the internship serves as a practical application of Python programming knowledge and provides an opportunity to develop technical proficiency through project-based learning. The combination of multiple applications allows different programming concepts to be practiced in meaningful contexts while improving logical thinking, debugging ability, code quality, and confidence in developing Python-based solutions.

4.2 Organization Profile

CodSoft is an innovative technology education and software development platform focused on empowering aspiring developers by providing opportunities to convert theoretical knowledge into practical technical expertise. The organization emphasizes project-based learning and skill development, enabling learners to gain practical exposure while working on application-oriented tasks. Its internship programs are designed to provide a structured learning environment in which participants can strengthen their technical abilities through hands-on development activities.

CodSoft provides virtual internship programs across technology-oriented areas and incorporates project-based learning as an important component of its training approach. The Python Programming internship is designed to provide learners with practical exposure to Python through a series of development tasks. Rather than focusing only on theoretical concepts, the internship encourages participants to implement working applications, understand programming logic, and improve their problem-solving capabilities through practical assignments.

The organization also provides a broader ecosystem of career-oriented resources. According to the internship material, CodSoft offers AI-driven career tools, including a resume builder, ATS score checking, AI interview coaching, career roadmap generation, portfolio-building support, and interview preparation resources. These services are intended to complement technical learning by helping aspiring developers prepare for professional opportunities.

A key aspect of CodSoft's approach is its emphasis on continuous learning and practical development. The organization recognizes that the technology industry evolves continuously and therefore promotes learning through practical projects and guidance. Its internship framework allows participants to work independently while also providing opportunities to seek guidance when required. The Python Programming internship specifically encourages interns to approach their assigned tasks with professional diligence and give appropriate attention to the quality of their work.

As part of the Python Programming internship, interns are required to complete a minimum of three assigned tasks for successful completion. The internship material also instructs participants to maintain a dedicated GitHub repository for their tasks and share the repository as part of the submission process. This approach provides practical experience not only in Python development but also in organizing, documenting, and presenting software projects through a version-controlled development platform.

Overall, CodSoft provides a project-oriented learning environment that combines technical development with career preparation. The Python Programming internship offers an opportunity to apply programming concepts to practical applications, strengthen software development skills, and gain experience in presenting completed work in a professional manner. Through this internship, the learner is exposed to the practical aspects of programming and encouraged to continuously improve technical knowledge and development practices.

4.3 Problem Statement

The rapid adoption of software-based solutions across different domains has increased the importance of developing practical programming skills and the ability to translate logical requirements into functional applications. While theoretical understanding of programming concepts provides a foundation for software development, practical implementation is essential for understanding how these concepts are applied to solve real-world problems. The primary challenge addressed during this internship is therefore to apply Python programming concepts in the development of functional, interactive, and logically structured applications.

The internship provides a set of application-oriented tasks that require the implementation of different programming concepts using Python. The objective is not limited to writing code that produces the desired output, but also involves designing programs with a clear structure, appropriate user interaction, reliable input processing, and maintainable logic. The assigned tasks include a To-Do List, Calculator, Password Generator, Rock-Paper-Scissors Game, and Contact Book, each representing a different type of programming problem.

The To-Do List addresses the problem of organizing and managing tasks through operations such as adding, updating, deleting, and tracking task status. The Calculator focuses on processing numerical inputs and performing arithmetic operations while handling invalid user inputs appropriately. The Password Generator addresses the requirement of generating randomized passwords using configurable character sets and user-defined length. The Rock-Paper-Scissors Game involves implementing decision-making logic, random computer selection, and interactive game flow. The Contact Book addresses structured management of contact information through operations such as adding, viewing, searching, updating, and deleting records.

A significant aspect of the problem is ensuring that these applications are not only functional but also capable of handling common input-related issues. Users may provide invalid menu choices, incorrect data types, empty values, or values outside an acceptable range. Therefore, appropriate input validation and exception-handling mechanisms need to be incorporated to improve program reliability and user experience. The projects also require logical decomposition into functions and manageable components so that the code remains readable and easier to maintain.

Thus, the overall problem addressed by the internship is the development of a collection of reliable and user-oriented Python applications while applying fundamental and intermediate programming concepts in a practical environment. The projects provide a structured opportunity to strengthen logical reasoning, programming implementation, debugging, validation, code organization, and software documentation skills. The resulting applications form a practical demonstration of the learner's ability to convert defined requirements into working Python-based solutions.

4.4 Project Objectives

The primary objective of this internship is to develop practical proficiency in Python programming by applying programming concepts to the development of functional and user-oriented applications. The project work focuses on strengthening programming fundamentals while providing practical exposure to application development, testing, debugging, and documentation.

The major objectives of the project are:

- To strengthen Python programming skills by applying variables, data types, operators, conditional statements, loops, functions, and data structures in practical applications.
- To improve problem-solving and logical thinking by analyzing requirements and converting them into structured and efficient programming solutions.
- To develop practical Python applications including a To-Do List, Calculator, Password Generator, Rock-Paper-Scissors Game, and Contact Book.
- To implement modular and maintainable code by dividing applications into appropriate functions and logical components, thereby improving readability and reusability.
- To implement input validation and error handling so that applications can respond appropriately to invalid inputs and common execution errors.
- To gain practical experience in data management through operations such as adding, viewing, searching, updating, and deleting information in applications such as the To-Do List and Contact Book.
- To apply Python modules and built-in functionality for implementing features such as randomization, password generation, and other supporting operations.
- To gain experience in software development practices including testing, debugging, code refinement, documentation, and systematic project development.
- To organize and present project work professionally by maintaining the internship tasks in a dedicated GitHub repository as specified in the internship guidelines.
- To bridge the gap between theoretical and practical learning by applying academic programming knowledge to the development of working software applications and building a foundation for further learning in software development and related technologies.

4.5 Scope of the Project

The scope of this internship project covers the design and development of multiple Python-based applications as part of the CodSoft Python Programming internship. The projects are intended to provide practical exposure to programming concepts and demonstrate their application in functional, interactive, and user-oriented software solutions.

The project scope includes the following applications:

- To-Do List – Development of a task management application that allows users to add, view, update, delete, and manage tasks according to their completion status.
- Calculator – Development of an interactive calculator capable of performing basic arithmetic operations while handling user input and common calculation errors.
- Password Generator – Development of a utility for generating randomized passwords based on user-defined requirements such as password length and character selection.
- Rock-Paper-Scissors Game – Development of an interactive game that allows a user to play against a computer-generated choice using predefined game rules and randomization.
- Contact Book – Development of a contact management application that allows users to add, view, search, update, and delete contact information using suitable Python data structures.

The project also covers the use of fundamental Python programming concepts such as variables, data types, conditional statements, loops, functions, lists, dictionaries, modules, and exception handling. These concepts are applied according to the requirements of individual applications to create structured and reliable programs.

An important part of the project scope is input validation and error handling. The applications are designed to respond appropriately to invalid choices, incorrect data types, missing information, and other common user-input issues. This helps improve the reliability and usability of the developed applications.

The scope also includes basic software development practices such as code organization, testing, debugging, documentation, and version control. The completed internship tasks are maintained in a dedicated GitHub repository, providing a structured platform for storing and presenting the project source code as required by the internship guidelines.

The projects are primarily developed as practical Python applications and provide a foundation for future enhancement. Possible extensions include graphical user interfaces, database integration, web-based implementation, advanced data management, and additional automation features. However, these extensions remain outside the primary scope of the current internship project.

4.6 Technologies and Tools Used

The development of the internship projects involved a combination of programming technologies and development tools used for writing, testing, organizing, and maintaining the Python applications. The major technologies and tools used are described below:

- Python 3.x – Python was used as the primary programming language for developing all the internship applications. Its simple syntax, extensive standard library, and support for modular programming made it suitable for implementing the assigned projects.
- Python Standard Library – Built-in modules and functionalities were used wherever required. Modules such as `random` and `string` can be utilized for features such as password generation and computer-generated choices in the Rock-Paper-Scissors application.
- Data Structures – Python lists and dictionaries were used for organizing and managing application data. These structures are particularly useful for applications such as the To-Do List and Contact Book.
- Functions and Modular Programming – Functions were used to divide the applications into smaller and manageable components. This improves code readability, reusability, testing, and maintainability.
- Exception Handling – Python exception-handling mechanisms were used to manage unexpected situations and invalid user inputs, helping improve the reliability of the applications.
- Integrated Development Environment (IDE) – A Python-compatible development environment such as Visual Studio Code was used for writing, editing, executing, and debugging the project source code.
- Command-Line Interface (CLI) – The applications were designed around a command-line interaction model, allowing users to provide inputs through menus and receive results directly through the terminal.

Overall, these technologies and tools provided the necessary environment for developing, testing, organizing, and presenting the Python applications completed as part of the internship.

4.7 System Architecture

The system architecture of the internship project represents the overall structure through which the Python applications receive user input, process the requested operation, manage application data, and generate the required output. Since the assigned projects are primarily command-line based, a simple and modular architecture is used. The architecture emphasizes clear separation between input handling, application logic, data management, and output generation.

The overall architecture consists of the following major components:

- **User Interface Layer** – The user interacts with the applications through a command-line interface. Menu options and prompts are provided to allow the user to select operations and enter the required information.
- **Input and Validation Layer** – User-provided data is collected and validated before processing. This layer checks whether the input is in the expected format and helps handle invalid choices, incorrect data types, empty values, or inappropriate inputs.
- **Application Logic Layer** – This is the main processing component of the system. It contains the functions and conditional logic responsible for performing the operations requested by the user. Different applications use different logic according to their requirements, such as arithmetic calculations, task management, password generation, game rules, or contact management.
- **Data Management Layer** – Structured Python data structures such as lists and dictionaries are used to organize and manage application information. For example, tasks can be represented as structured records in the To-Do List, while contact details can be maintained using suitable key-value structures in the Contact Book.
- **Utility and Processing Components** – Supporting Python functionality and standard-library modules are used where required. For example, randomization can be used for generating passwords and producing computer choices in the Rock-Paper-Scissors application.
- **Output Layer** – After processing the requested operation, the application generates an appropriate response and displays the result through the command-line interface. Clear output messages help the user understand the result of each operation.

The general flow of the system can be represented as:

User → Command-Line Interface → Input Validation → Application Logic → Data/Utility Processing → Output

The architecture follows a modular approach in which individual functions are responsible for specific operations. This reduces unnecessary complexity and makes the applications easier to understand, test, debug, and modify.

4.8 Methodology

The methodology adopted for the internship project follows a systematic and practical approach to software development. The assigned Python programming tasks were completed by understanding the requirements, planning the program structure, implementing the required functionality, testing the applications, and refining the code. This approach helped in applying programming concepts in a structured manner and developing applications that are reliable, understandable, and easy to maintain.

- **Requirement Analysis** – The requirements of each assigned task were first studied to understand the purpose of the application, required features, user inputs, processing operations, and expected outputs. This helped in defining the functionality that needed to be implemented in each project.
- **Planning and Program Design** – After analyzing the requirements, the overall flow of each application was planned. Appropriate functions, conditions, loops, data structures, and Python modules were selected based on the requirements. The program was divided into logical components to maintain a clear and organized structure.
- **Implementation** – The applications were developed using Python and its core programming features. Concepts such as variables, data types, operators, conditional statements, loops, functions, lists, dictionaries, and modules were used to implement the required functionality. Each project was developed according to its individual requirements while maintaining a consistent programming approach.
- **Input Handling and Validation** – User interaction was an important part of the applications. Inputs were collected through the command-line interface and checked before processing. Validation was incorporated to handle incorrect choices, unsuitable values, invalid data types, and other common input-related situations.
- **Testing** – The developed applications were tested with different sets of inputs to verify their functionality. Normal test cases were used to check expected operations, while invalid and boundary inputs were also considered to identify possible issues in program execution.
- **Debugging** – Errors encountered during development and testing were analyzed and corrected systematically. Both logical and runtime issues were reviewed to ensure that the applications produced appropriate results under different conditions.
- **Code Refinement** – After successful testing, the programs were reviewed to improve code readability, organization, and maintainability. Repeated logic was reduced where possible, functions were structured appropriately, and user messages were kept clear and meaningful.

The overall methodology follows an iterative cycle in which requirements are understood first, followed by design, implementation, testing, debugging, and refinement.

4.9 Expected and Observed Outcomes

The internship project was designed to provide practical exposure to Python programming through the development of multiple application-oriented tasks. The expected outcomes were focused on improving programming knowledge, problem-solving ability, application development skills, and familiarity with basic software development practices. The observed outcomes were evaluated through the implementation, testing, and refinement of the developed applications.

Expected Outcomes

- To develop a stronger practical understanding of Python programming concepts and their application in software development.
- To improve logical thinking and problem-solving skills by converting defined requirements into functional Python applications.
- To develop working applications such as the To-Do List, Calculator, Password Generator, Rock-Paper-Scissors Game, and Contact Book.
- To gain practical experience with functions, conditional statements, loops, lists, dictionaries, modules, and exception handling.
- To understand the importance of input validation, error handling, testing, and debugging while developing interactive applications.
- To improve the ability to organize source code into logical and maintainable components.
- To gain experience in maintaining and presenting project work through a dedicated GitHub repository as required by the internship guidelines.

Observed Outcomes

- The implementation of the assigned applications provided practical experience in applying Python programming concepts beyond theoretical learning.
- The development of different applications improved understanding of program flow, functions, data handling, conditional logic, and user interaction.
- Testing with different inputs helped identify programming errors and improved the ability to debug and refine the applications.
- Input validation and error handling improved the reliability of the programs and provided a better understanding of handling unexpected user inputs.

Overall, the expected objectives of the internship were aligned with the observed learning outcomes. The project-based approach provided practical experience in Python development and helped strengthen programming logic, debugging, code organization, and

technical problem-solving skills. The developed applications also established a foundation for further learning and enhancement of Python-based software solutions.

4.10 Internship Offer letter



INTERNSHIP OFFER LETTER

Date : 24/08/2026

ID: BY26RY232903

Dear **Prakhar Tripathi**,

We would like to congratulate you on being selected for the “**Python Programming**” virtual internship position with “**CodSoft**”. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **25 August 2026 to 25 September 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!



ISO Number - TSNUK79053

Sincerely

Ruman Arshad
Program Manager



MSME Regd



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